

Non-asymptotic confidence bands on the probability an individual benefits from treatment (PIBT)

Gabriel Ruiz, Oscar Madrid Padilla

Department of Statistics

University of California, Los Angeles

20 May 2022

Introduction

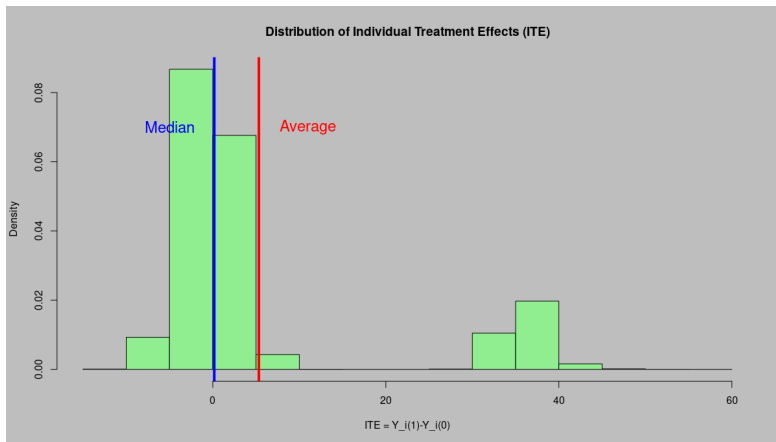


Figure: The mean as a summary of the individual treatment effects.

Introduction

In Ruiz and Padilla (2022), we seek **closed-form**, non-asymptotic confidence bands on the **inestimable** probability an individual benefits from treatment (**PIBT**):

- ① $\Pr(Y_i(1) - Y_i(0) > \delta)$
- ② $\Pr(Y_i(1)/Y_i(0) > \delta)$ (positive outcomes)
- ③ $\Pr(Y_i(1) - Y_i(0) > \delta | X_i = x).$

The challenge

Probability an individual benefits from treatment:

$$\Pr(Y_i(1) - Y_i(0) > \delta).$$

- ① Cannot be estimated.
- ② Not even with matching.

The challenge

Probability an individual benefits from treatment:

$$\theta^L(\delta) \leq \Pr(Y_i(1) - Y_i(0) > \delta) \leq \theta^U(\delta).$$

- ① We have to bound this quantity, then estimate those bounds.
- ② Two forms of uncertainty: The best possible bounds, and **our estimation error for these bounds.**

- Extension of existing **plug-in estimation** framework (Fan and Park, 2010).
- **Distribution-free** bounds on CDF of difference (any possible dependence). (Williamson and Downs, 1990; Makarov, 1982).
- Non-parametric Empirical CDF deviation inequality (Dvoretzky et al., 1956; Massart, 1990).
- Neyman-Rubin causal model; marginal and conditional ignorability.

Insight 1: Practical Interpretation

With $(1 - \alpha) \times 100\%$ confidence, PIBT is contained between:

[Estimator for lower bound on PIBT] – [Margin of error for PIBT bounds]

and

[Estimator for upper bound on PIBT] + [Margin of error for PIBT bounds].

Insight 2: Power Analysis For Randomized Controlled Trial

- Margin of error for PIBT bounds = $\left(\frac{\log(4/\alpha)}{2}\right)^{\frac{1}{2}} \left(n_0^{-\frac{1}{2}} + n_1^{-\frac{1}{2}}\right)$ with $(1 - \alpha) \times 100\%$ confidence.

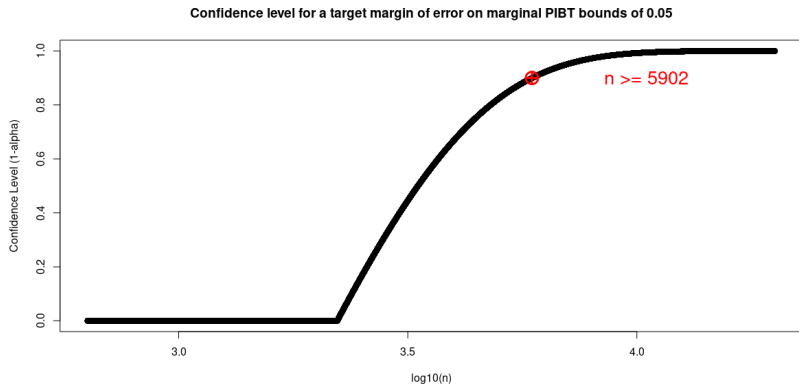


Figure: Power analysis based on our *Theorem 1* (Ruiz and Padilla, 2022). Here, $n_0 = n_1$, but can deal with differing sample sizes.

Insight 3: Heterogeneity in PIBT

- **Simultaneous inference** at all strata x , thresholds δ :

$$\sup_{x, \delta} \left\{ \left| \hat{\theta}^L(\delta, x) - \theta^L(\delta, x) \right| \vee \left| \hat{\theta}^U(\delta, x) - \theta^U(\delta, x) \right| \right\}.$$

- Key: **impute** draws from conditional outcome distribution

$$\hat{\mu}_w(x) + \{Y_i(w) - \hat{\mu}_w(X_i)\} \stackrel{d}{\approx} Y_i(w) | X_i = x.$$

- Target Confidence Level **Correction**:

$$1 - \alpha - \Pr \left(\bigvee_{w=0,1} \sup_x |\hat{\mu}_w(x) - \mu_w(x)| > t_w \right).$$

Insight 3: Heterogeneity in PIBT

At sample size n , what is the uniform Margin of Error on efficacy bounds at the 90% confidence level?

Number of Features $\color{red}\bullet$ $p = 2$ $\color{olive}\bullet$ $p = 5$ $\color{teal}\bullet$ $p = 10$ $\color{blue}\bullet$ $p = 25$ $\color{violet}\bullet$ $p = 50$

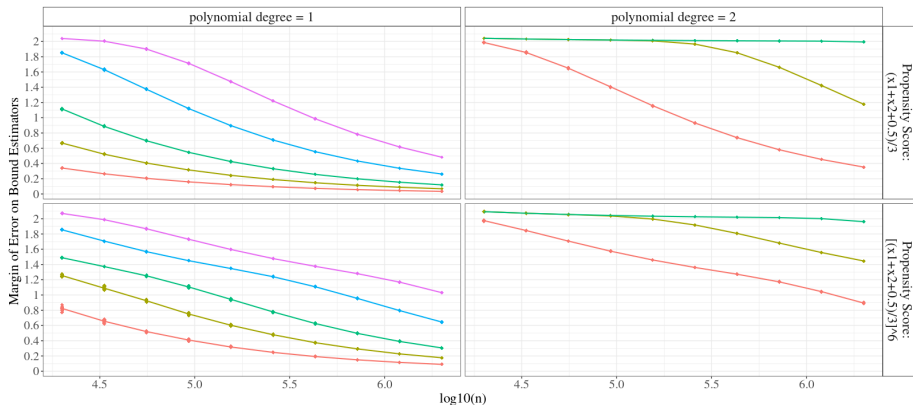


Figure: Power analysis based on our *Theorem 3* applied to canonical linear model (Ruiz and Padilla, 2022).

Conclusion

- We provide closed form high probability concentration inequalities for our estimation of bounds on $\Pr(Y_i(1) - Y_i(0) > \delta)$.
 - i.e. a closed-form “Margin of Error” that allows for a power analysis.
- We extended our theory to case with pre-treatment covariates.
 - A similar power analysis is possible.

Acknowledgements

Qing, Oscar, UCLA causal inference reading group

NSF-GRFP DGE-1650604

Time to event outcomes?

References I

- Dvoretzky, A., Kiefer, J., and Wolfowitz, J. (1956). Asymptotic Minimax Character of the Sample Distribution Function and of the Classical Multinomial Estimator. *The Annals of Mathematical Statistics*, 27(3):642 – 669.
- Fan, Y. and Park, S. S. (2010). Sharp bounds on the distribution of treatment effects and their statistical inference. *Econometric Theory*, 26(3):931–951.
- Makarov, G. D. (1982). Estimates for the distribution function of a sum of two random variables when the marginal distributions are fixed. *Theory of Probability & Its Applications*, 26(4):803–806.
- Massart, P. (1990). The Tight Constant in the Dvoretzky-Kiefer-Wolfowitz Inequality. *The Annals of Probability*, 18(3):1269 – 1283.
- Ruiz, G. and Padilla, O. H. M. (2022). Non-asymptotic confidence bands on the probability an individual benefits from treatment (pibt).
- Williamson, R. C. and Downs, T. (1990). Probabilistic arithmetic. i. numerical methods for calculating convolutions and dependency bounds. *International Journal of Approximate Reasoning*, 4(2):89–158.